PHYSICS-AWARE MACHINE LEARNING FOR BIO-OIL PYROLYSIS: AN HONEST BENCHMARK AND A PYROLYSIS ADVISOR FOR HIGHER YIELD AND LOWER OXYGEN

A PROJECT REPORT

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BONA FIDE CERTIFICATE

This is to certify that the project report entitled "PHYSICS-AWARE MACHINE LEARNING FOR BIO-OIL PYROLYSIS: AN HONEST BENCHMARK AND A PYROLYSIS ADVISOR FOR HIGHER YIELD AND LOWER OXYGEN" submitted by Mohalraj Shivanethran (CB.AI.U4AIM25020), Pranesh Subramanian (CB.AI.U4AIM25030), Srinath Balakrishnan (CB.AI.U4AIM25040) and Umasuthan Palaniappan (CB.AI.U4AIM25041) in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Artificial Intelligence and Data Science Engineering is a bona fide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Coimbatore.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

I express my sincere gratitude to my faculty guide, Dr. Akhil Mohan, for his continuous guidance, technical insight and encouragement throughout this project, and for directing this work towards a problem of genuine relevance to his own research on bio-oil separation strategies.

I thank the Department of Artificial Intelligence and Data Science, Amrita School of Engineering, Coimbatore, and Amrita Vishwa Vidyapeetham for providing the academic environment, computing resources and the course framework (24AIM201, Mechanics and Fluidized Systems) within which this project was carried out.

I also acknowledge the authors of the two public datasets used in this study — Zhang et al. (2022) and Pascarella et al. (2025) — whose openly shared literature data made this analysis possible, and whose own cautionary notes on data sparsity motivated the honest-evaluation approach taken here.

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ABSTRACT

Machine learning (ML) is increasingly used to predict bio-oil yield and quality from biomass composition and pyrolysis conditions, and published models often report R^2 above 0.75. This project re-examines two public literature datasets: bio-oil O/C ratio (Zhang et al. 2022, 223 clean rows) and bio-liquid yield (Pascarella et al. 2025, 1,046 rows). It is shown that the usual random train/test split lets rows from the same biomass appear in both training and testing, which inflates scores. For yield, R^2 falls from 0.66 (random split) to 0.23 when rows are grouped by exact composition, and to −0.18 when hidden near-duplicate rows of the same biomass are also grouped. For O/C, R^2 falls from 0.63 to 0.23.

Physics-aware models were then built using heat-transfer, kinetic and molar-ratio features, physical constraints, and honest grouped validation. Physics features raised the unseen-biomass R^2 for O/C in all three model types tested (for example GPR from −0.14 to 0.23). The final models reproduce known physics: yield peaks near 525 °C and O/C falls from 0.28 to 0.21 between 350 and 700 °C. These models power a Pyrolysis Advisor that recommends reactor settings balancing high yield and low oxygen, flags biomasses outside its knowledge, and improves its predictions with one or two real experiments. A supporting FastAPI/React web application, built on an earlier single-dataset iteration of this work, is also documented as a demonstration of deployment engineering. The work addresses the lack of transferable, physics-consistent ML noted in Dr. Akhil Mohan's review of bio-oil separation strategies, and prepares lower-oxygen oil for the separation stage.

Keywords: bio-oil, pyrolysis, machine learning, data leakage, physics-informed features, Gaussian process regression, decision support.

TABLE OF CONTENTS

(Right-click the table above and select "Update Field" → "Update entire table" in Microsoft Word to populate page numbers after any further edits.)

LIST OF FIGURES

Figure 6.1: Test-set R^2 under different testing methods (physics inputs)

Figure 6.2: O/C on unseen biomass: raw inputs vs. physics inputs

Figure 7.1: Average model response to temperature, with kinetic tar selectivity for comparison

Figure 7.2: Yield error for a new biomass with 0, 1 or 2 real experiments

Figure 7.3: Held-out biomasses; the circled point is the single experiment used for calibration

Figure 7.4: Advisor output for a pine-like wood (C 50, H 6.1, O 43.5, N 0.2 wt%; particle 0.5 mm)

LIST OF TABLES

Table 2.1: Summary of six literature datasets and published ML models for bio-oil pyrolysis

Table 3.1: Summary of the two datasets used in this study, after cleaning

Table 3.2: Row-level cleaning actions applied to D1b and D3

Table 4.1: Physics-informed features engineered from raw inputs

Table 5.1: Model families and hyperparameters used

Table 5.2: Three train/test protocols compared in Part A

Table 6.1: Full honest-benchmark results: random split vs. grouped-by-biomass split

Table 7.1: Final model selection: level/shape yield model vs. alternatives

Table 7.2: Final O/C model selection: GPR, constrained XGBoost and their blend

Table 7.3: Case studies: three biomasses withheld entirely from training

Table 7.4: Pyrolysis Advisor recommendations for four example feedstocks

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

A — Pre-exponential (frequency) factor in an Arrhenius rate expression

ASE — Amrita School of Engineering

C, H, O, N — Mass percentages of carbon, hydrogen, oxygen and nitrogen in the feedstock (ultimate analysis)

Ea — Activation energy

FeedID — Internal grouping identifier assigned to rows sharing the same (near-duplicate-corrected) biomass composition

GPR — Gaussian Process Regression

HHV — Higher Heating Value

HR — Heating rate (°C/min)

k — Reaction rate constant

MAE — Mean Absolute Error

MedAPE — Median Absolute Percentage Error

ML — Machine Learning

O/C, H/C — Molar oxygen-to-carbon and hydrogen-to-carbon ratios

PS — Feedstock particle size (mm)

R — Universal gas constant, 8.314 J mol^{-1} K^{-1}

R^2 — Coefficient of determination

RF — Random Forest

S_tar — Kinetic tar selectivity (share of biomass forming tar/oil rather than gas or char)

T — Pyrolysis (process) temperature (°C)

TSI — Thermal Severity Index

XGBoost — Extreme Gradient Boosting

θ — Heating-regime parameter ($\log_{10}$ of conduction time over heating time)

CHAPTER 1: INTRODUCTION

1.1 Background

Pyrolysis heats biomass to 400–600°C without oxygen to produce bio-oil, char and gas. Crude bio-oil carries 31–52 wt% oxygen and 15–30 wt% water, making it acidic, unstable and low in heating value (Mohan et al. 2026), so it must be upgraded or separated before use. Dr. Akhil Mohan's review of bio-oil separation strategies identifies phase separation as the most promising upgrading route, and notes (Section 8.5) that no published work yet integrates ML with the separation stage, because separation datasets do not exist.

This project therefore targets the stage immediately upstream of separation: the pyrolysis reactor. A reactor operated to produce inherently lower-oxygen bio-oil reduces the solvent, adsorbent and energy demand of the downstream separation stage — predicting and recommending conditions that trade off yield against oxygen content is a natural precursor to the separation problem Dr. Mohan's review identifies.

1.2 Problem Statement

- No separation data — public pyrolysis datasets cover only the reaction stage, not the separation stage.
- Over-optimistic ML — 65% of published pyrolysis ML studies validate using random cross-validation, which inflates accuracy when related rows appear in both train and test sets (da Costa et al. 2027).
- Little physics — most studies feed raw numbers into black-box regressors with no physical grounding and no basis for trusting a prediction on an unfamiliar feedstock.

1.3 Objectives

- Part A: quantify how strongly the testing protocol inflates reported ML accuracy for bio-oil yield and O/C, using two public datasets.
- Part B: build physics-aware models for yield and O/C that are honestly validated (on genuinely new biomasses) and that respect known pyrolysis physics.

- Package the models into a Pyrolysis Advisor that recommends reactor conditions, flags unfamiliar feedstocks, and improves with 1–2 calibration experiments; and, as supporting engineering, a deployable web application.

CHAPTER 2: LITERATURE REVIEW

Six recent datasets and their associated studies were surveyed to situate this project (Table 2.1). A common pattern is visible: reported accuracy is consistently high (R^2 typically 0.75–0.97), models use little explicit process physics, and none address the separation stage.

Table 2.1: Summary of six literature datasets and published ML models for bio-oil pyrolysis

ID	Source	Data	Predicts	Best model	R a
D1a	Tang et al. 2020, Energy & Fuels 34(9)	~262 rows	Yield, H in oil	Random Forest	R^2 0
D1b ★	Zhang et al. 2022, Fuel 312:122812	232 O/C rows	Yield, viscosity, HHV, H/C, O/C	Random Forest	All
D2	Ortiz-Alvarez et al. 2022, Energy Sources A	329 rows, 34 biomasses	Yield	Neural network	R = 0.
D3 ★	Pascarella et al. 2025, Fuel 394:135000	1,137 rows, ~160 studies	Bio-liquid yield	XGBoost	R^2 0.
D4a	Zhao, Xia et al. 2025, J. Energy Inst.	122 rows	Co-pyrolysis yield	Deep neural network	
D4b	Xia et al. 2026, J. Energy Inst.	109 rows	Oxygenated compounds	Deep neural network	

Two sources were used directly in this project. Zhang et al. (2022, Fuel) compiled bio-oil yield, viscosity, HHV, H/C and O/C from biomass composition and conditions, modelled with Random Forest on a 70/30 random split; O/C was their weakest target (errors often above 20%). This dataset (D1b) supplies the O/C target here. Pascarella et al. (2025, Fuel) assembled the largest yield dataset surveyed — 1,137 rows from ~160 source studies (Zenodo 10.5281/zenodo.19205977) — and explicitly caution that the data is sparse and recommend physics-informed modelling, a recommendation this project takes up directly (Chapters 4, 7). This dataset (D3) supplies the yield target.

Methodological grounding also came from: da Costa et al. (2027), whose survey finding that 65% of studies use random cross-validation motivated the grouped-validation protocol (Chapter 5); Chan, Kelbon & Krieger (1985), whose wood-pyrolysis kinetic scheme underlies the tar-selectivity feature (Chapter 4); Ren, Wu

& Weng (2023), on monotonic physics-informed constraints; and Aye et al. (2026), whose thermal-severity-index formulation is adapted in Chapter 4.

Relative to these six studies, this project's contribution is threefold: a quantitative demonstration of how much random-split testing inflates accuracy on these exact datasets, including a previously undocumented near-duplicate problem in D3; a physics-informed feature and constraint layer built from heat-transfer and kinetic first principles; and an explicit calibration mechanism using one or two real experiments on a new biomass.

CHAPTER 3: DATASETS AND DATA PREPARATION

Table 3.1: Summary of the two datasets used in this study, after cleaning

ID	Source	Rows used	Biomasses	Inputs → Output
D1b	Zhang et al. 2022, Fuel 312:122812	223	122	Feed C, H, O, N, particle size, heating rate, T → bio-oil O/C
D3	Pascarella et al. 2025, Fuel 394:135000	1,046	118	Same (+ ash, VM, FC, cellulose, hemicellulose, lignin) → yield (wt%)

3.1 Cleaning — D1b (O/C)

320 raw rows. Gas_yield/Char_yield (empty) dropped; 6 numeric columns corrected from stray text (non-breaking spaces, dashes); 4 exact duplicates removed; 13 physically implausible rows excluded (H%>15 or N%>30 or particle size>20mm suggesting corruption; C%≈15% suggesting a column shift; N%≈20% transcription errors; char-like feed at <200°C or any row <300°C with non-char composition). Rows were grouped by identical (C,H,O,N) into 122 FeedIDs. The final modelling set keeps only complete, unflagged rows: 223.

3.2 Cleaning — D3 (Yield)

1,057 raw rows; 11 exact duplicates removed (1,046 remain). Grouping by exact composition gives 651 apparent biomasses — but every composition value carries tiny decimal differences for what is evidently the same biomass (e.g. C = 49.2073% vs 49.2074%), an artefact of the source compilation. Hierarchical single-linkage clustering of (C,H,O,N,Ash) at a 0.2 wt% threshold reduces this to 118 true biomasses, stable across the 0.1–0.3 wt% range (120–116 groups). This distinction matters directly for Chapter 6: grouping by exact value still leaks the same biomass across folds under its differing decimals.

Table 3.2: Row-level cleaning summary

Dataset	Rows raw	Duplicates	Other exclusions	Rows retained	Biomasses
D1b (O/C)	320	4	13 (implausible)	223	122
D3 (Yield)	1,057	11	0	1,046	118 (near-dup corre

CHAPTER 4: PHYSICS-INFORMED FEATURE ENGINEERING

Rather than feeding raw composition/process columns directly into a statistical learner, these are first converted into physically meaningful quantities derived from heat-transfer theory, reaction kinetics and stoichiometry (Table 4.1), so the model is told in part how pyrolysis behaves physically rather than relying purely on correlation from a small, heterogeneous sample.

Table 4.1: Physics-informed features engineered from raw inputs

Feature	Formula	Physical meaning
Feed O/C, H/C (molar)	(O/16)/(C/12), (H/1.008)/(C/12)	Oxygen / hydrogen atoms per carbon atom in the feedstock
Feed effective H/C	(H/1.008 − 2·O/16)/(C/12)	Hydrogen remaining after O is consumed forming water
Heating regime θ	$\log_{10}$(t_cond/t_heat); t_cond=r^2/α, t_heat=(T−25)/HR	Whether the particle heats evenly (θ<0) or lags at its core
Tar selectivity S_tar	k_tar/(k_gas+k_tar+k_char), Arrhenius (Chan et al. 1985)	Fraction of biomass predicted to form tar/oil vs. gas or char
Pyrolysis number	$\log_{10}$(reaction time / conduction time)	Reaction-rate-limited vs. heat-transfer-limited process
Thermal severity index	$\log_{10}$(T·ln(HR+1)/PS)	Combined measure of process thermal aggressiveness (Aye et al. 2026)
log HR, log PS, T, N	As given, log-transformed where skewed	Process conditions on a scale suited to the regressors

4.1 Reaction Kinetics: Tar Selectivity

Based on the classical three-reaction wood-pyrolysis scheme of Chan, Kelbon & Krieger (1985): gas, tar and char each follow a first-order Arrhenius law, k_gas=$1.43\times10^4\cdot\exp(-88{,}600/RT)$, k_tar=$4.13\times10^6\cdot\exp(-112{,}700/RT)$, k_char=$7.38\times10^5\cdot\exp(-106{,}500/RT)$ (T in K, R=8.314 J $mol^{-1}K^{-1}$). Tar selectivity S_tar = k_tar/(k_gas+k_tar+k_char) rises with temperature because the tar channel has the highest activation energy of the three — a direct kinetic explanation for why moderate-to-high temperatures favour liquid oil, given to the model as a feature rather than left to correlation alone.

4.2 Heat Transfer: Heating Regime

The conduction timescale $t_{cond}=r^2/\alpha$ ($\alpha=1.0\times10^{-7}$ m²/s, representative biomass thermal diffusivity) is compared with the heating timescale $t_{heat}=(T-25)/HR$ to give $\theta=\log_{10}(t_{cond}/t_{heat})$. Negative θ means the particle heats faster than the reactor programme changes (evenly-heated, kinetically-controlled); every experiment in D3 has $\theta<0$ (max -0.86) at a median heating rate of only 25°C/min, confirming the compiled data is dominated by slow pyrolysis of small, evenly-heated particles (Chapter 7) and explaining why particle-size effects are small and noisy in the final models.

4.3 Physical Constraints

Beyond input features, physical reasoning is built directly into the O/C model as monotonicity constraints during training (Chapter 5): predicted O/C can never increase with process temperature, thermal severity or tar selectivity, and can never decrease as feedstock O/C increases — preventing predictions that contradict elementary pyrolysis chemistry even outside the dense training region.

CHAPTER 5: MODEL DEVELOPMENT AND VALIDATION PROTOCOL

5.1 Model Families

Table 5.1: Model families and hyperparameters used

Model	Configuration	Implementation
Mean (no model)	Predicts training-set mean for every input	— (baseline)
Random Forest	400 trees, min 2 samples/leaf, 60% feature subsampling	scikit-learn
XGBoost	500 trees, depth 3, lr 0.03, 80% subsampling, min child weight 5, L2=5.0	xgboost
GPR	Matérn kernel (ν=1.5) + white noise, standardised, target normalised	scikit-learn

5.2 Three Validation Protocols

Table 5.2: Three train/test protocols compared in Part A

Protocol	Mechanism	Risk / Rationale
Random split	80/20 random split, 10 repeats with different seeds.	Same biomass can appear in both train and test — common literature practice (Table 2.1).
Grouped, exact values	All rows with identical (C,H,O,N) kept together (5 folds × 5 repeats).	Still vulnerable to the hidden near-duplicate problem (Section 3.2).
Grouped by biomass	All rows of a biomass, incl. near-duplicates, kept together (5×5; 10 splits for GPR yield).	Simulates a genuinely new biomass — the real deployment scenario.

5.3 Physics-Constrained O/C Model

An XGBoost regressor with monotonicity constraints (predicted $\log_{10}$(O/C) non-increasing in T, thermal severity index and tar selectivity; non-decreasing in feed O/C) is blended 50/50 with an unconstrained GPR on the same features. The GPR alone under-responds to temperature while constrained XGBoost alone is higher-variance across folds; the blend generalised best to unseen biomasses (Chapter 7).

5.4 Level-Plus-Shape Yield Model

72% of D3's yield variance lies between biomasses rather than within one biomass across conditions (Section 6.3), so a plain model under-uses the reactor-condition

features. The final design separates a level GPR (biomass-composition features only, predicting a biomass's characteristic yield) from a shape XGBoost (process-condition features only, predicting deviation from that biomass's own average); the final prediction sums the two. This separates “which biomass” from “how yield responds to conditions,” which Chapter 7 shows is necessary for useful, condition-responsive advice.

5.5 Uncertainty, Calibration and Applicability

- Honest uncertainty: 80% intervals derived empirically from grouped out-of-fold errors on unseen biomasses, not a parametric assumption.
- Calibration: with 1–2 real experimental points, the prediction is shifted to match the measured offset, then used for the remaining conditions.
- Applicability check: a candidate's composition (feed O/C, H/C, N), z-score standardised, is compared via k-d tree nearest-neighbour search against training biomasses; flagged out-of-domain if its distance exceeds the 95th-percentile training nearest-neighbour distance.

5.6 Pyrolysis Advisor Search and Scoring

Given composition and particle size, the advisor evaluates 156 settings (26 temperatures 400–650°C × 6 heating rates 5–300°C/min), predicts yield and O/C for each, identifies the Pareto-optimal frontier, and scores each setting with 50/50 weighted normalised yield and inverted O/C to select the top recommendation.

CHAPTER 6: RESULTS AND DISCUSSION — PART A: HONEST BENCHMARK

This chapter quantifies how strongly validation protocol (Table 5.2) inflates reported accuracy. Results use raw and physics-informed features interchangeably as noted; Chapter 7 isolates the physics-feature contribution.

Table 6.1: Full honest-benchmark results (physics-informed features)

Target	Model	Test protocol	R^2	MAE (wt% yield)
Yield	Random Forest	Random split	0.69	3.8
Yield	XGBoost	Random split	0.66	4.2
Yield	GPR	Random split	0.60	4.3
Yield	Random Forest	Grouped, exact values	0.20	6.2
Yield	XGBoost	Grouped, exact values	0.23	6.3
Yield	Random Forest	Grouped by biomass	−0.25	8.7
Yield	XGBoost	Grouped by biomass	−0.18	8.3
Yield	GPR	Grouped by biomass	0.01	7.7
O/C (log scale)	Random Forest	Random split	0.52	—
O/C (log scale)	XGBoost	Random split	0.51	—
O/C (log scale)	GPR	Random split	0.63	—
O/C (log scale)	Random Forest	Grouped by biomass	0.19	—
O/C (log scale)	XGBoost	Grouped by biomass	0.14	—
O/C (log scale)	GPR	Grouped by biomass	0.23	—

For D1b, grouping by exact value and by biomass coincide (no hidden decimal noise); the grouped-by-biomass row is the only grouped result for O/C.

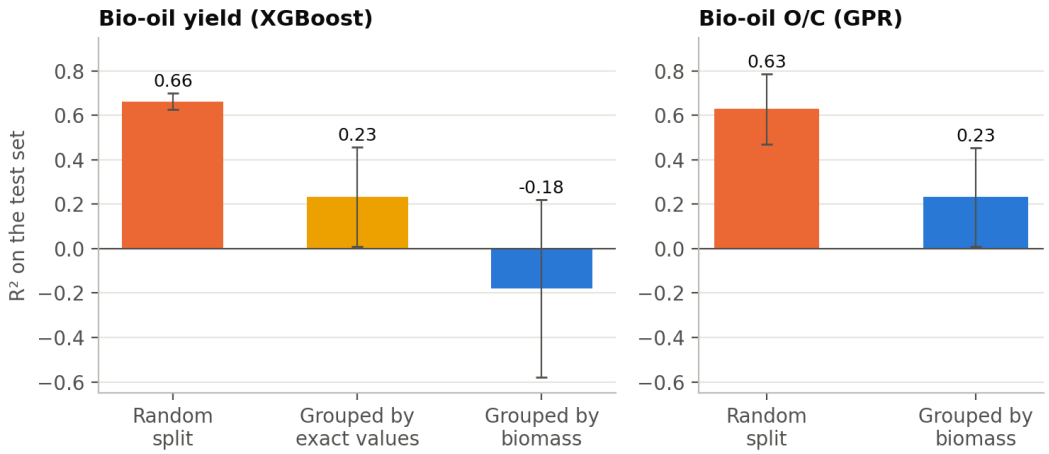


Figure 6.1: Test-set R² under different testing methods (physics inputs).

6.1 Random Splits Inflate Accuracy

Yield R² falls from 0.66 (random split) to −0.18 on a genuinely unseen biomass — no better than guessing the average. O/C falls from 0.63 to 0.23. These random-split figures match the 0.75–0.97 range surveyed in Chapter 2, supporting that much published accuracy reflects recognising already-seen biomasses rather than genuine generalisation.

6.2 Hidden Near-Duplicates Defeat Simple Grouping

Grouping D3 by exact value alone still gives yield R² = 0.23, far above the −0.18 obtained once near-duplicates are correctly clustered (Section 3.2) — a data-compilation artefact subtle enough to defeat a naive attempt at honest validation.

6.3 Why Yield Does Not Transfer

72% of D3's yield variance lies between biomasses rather than within one across conditions. D3 compiles ~160 source studies, each with its own lab, reactor and collection method; no recorded column distinguishes these systematic inter-lab effects from genuine biomass differences, so no model can learn what isn't in any input — a fundamental ceiling on achievable accuracy with this dataset.

6.4 Physics Features Help Most Where It Matters

On the honest grouped-by-biomass protocol, physics-informed features improved O/C prediction for every model tested; GPR rose from R²=−0.14 (raw) to 0.23

(physics) — from worse than baseline to usable. This directly supports Pascarella et al.'s (2025) own recommendation for physics-informed approaches on sparse compiled data.

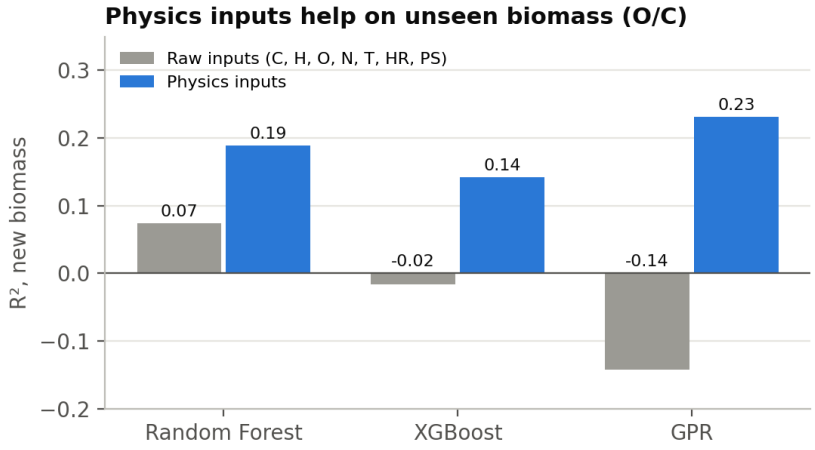


Figure 6.2: O/C on unseen biomass — raw vs. physics inputs.

CHAPTER 7: RESULTS — PART B: PHYSICS-AWARE MODELS AND THE ADVISOR

7.1 Final Model Selection

Table 7.1 compares yield candidates (grouped-by-biomass). "Shape R^2" measures how well a model captures the within-biomass response to conditions — what the Advisor needs. A plain GPR achieves shape $R^2 \approx 0.01$ (ignores conditions); level-plus-shape was selected for its shape $R^2 = 0.12$.

Table 7.1: Final yield model selection (grouped by biomass)

Model	R^2	MAE (wt%)	Shape R^2
Mean only	−0.09	8.21	0
GPR only	0.01	7.70	0.01
GPR + XGBoost average	0.07	7.48	0.07
Level + shape — final	0.07	7.50	0.12

Table 7.2: Final O/C model selection (grouped by biomass)

Model	R^2 (log)	Typical error (MedAPE)
GPR only	0.23 (log)	27.7% typical
XGBoost, physics constraints	0.14 (log)	37.5% typical
Blend — final	0.26 (log)	32.1% typical

7.2 Physics Consistency Checks

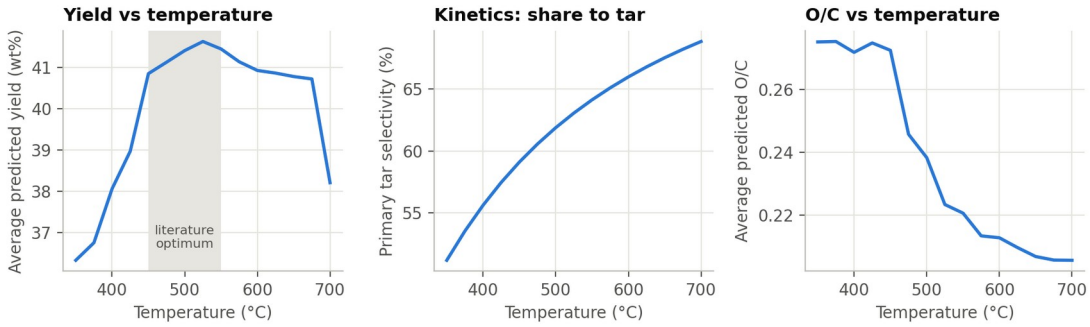


Figure 7.1: Average model response to temperature, with kinetic tar selectivity.

- Yield peaks at 525°C (41.6 wt%, rising from 36.3 at 350 and falling to 38.2 at 700) — matching the literature's 450–550°C optimum.
- Tar selectivity rises monotonically (51%→69% across 350–700°C), so the yield fall above ~600°C is consistent with secondary cracking of hot vapours into gas — a two-step behaviour reproduced from data, not told to the model explicitly.
- O/C falls 0.28→0.21 (350–700°C), consistent with stronger deoxygenation at higher severity, and is guaranteed never to rise with T by the monotonicity constraint.
- Every D3 experiment has $\theta<0$ (slow, evenly-heated pyrolysis; median HR 25°C/min), so particle-size effects are small and noisy.

7.3 Uncertainty and Calibration

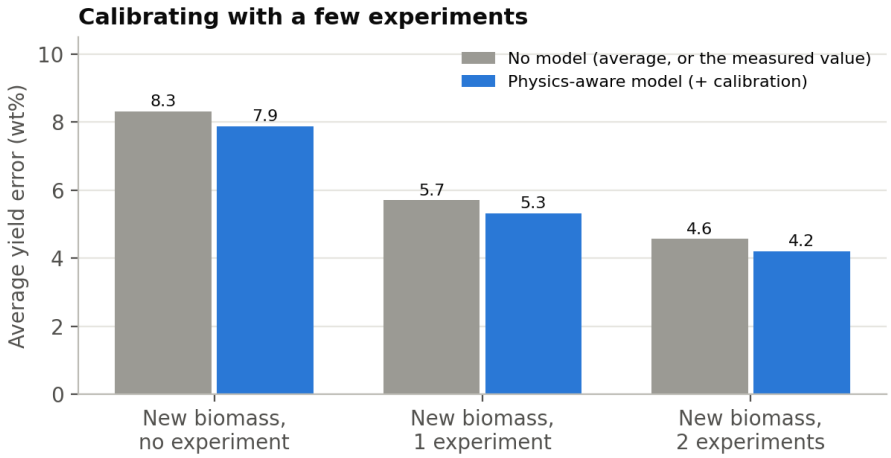


Figure 7.2: Yield error for a new biomass with 0, 1 or 2 real experiments.

80% intervals for a brand-new biomass: yield ±12.6 wt%, O/C within a factor of 2.2. One calibration experiment cuts yield error from ~8 to 5.3 wt%; two bring it to 4.2 wt% — beating simple reuse of the measured value (5.7, 4.6 wt%) because the model adds learned condition-response. For O/C the reverse held (reuse: log error 0.087 vs. model 0.104), since O/C varies little within a biomass in D1b.

7.4 Case Studies: Three Unseen Biomasses

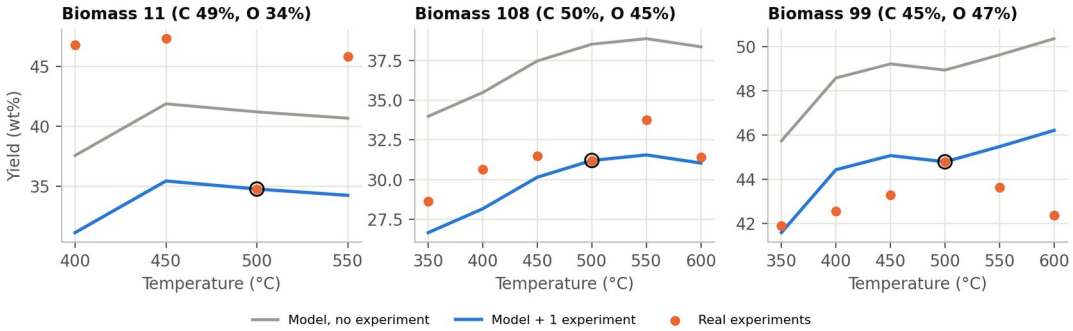


Figure 7.3: Held-out biomasses; the circled point is the calibration experiment.

Table 7.3: Case studies — biomasses withheld entirely from training

Biomass	Best T (real)	Best T (model)	Error, no exp.	Error, 1 exp.
11 (C 49%, O 34%)	450°C	450°C ✓	6.6 wt%	13.0 wt%
108 (C 50%, O 45%)	550°C	550°C ✓	5.6 wt%	1.7 wt%
99 (C 45%, O 47%)	500°C	600°C	6.0 wt%	1.9 wt%

The model found the correct optimum temperature for 2 of 3 biomasses with zero experiments; calibration cut error below 2 wt% for biomasses 108 and 99. Biomass 11's calibration point was an outlier relative to its neighbours (likely a data inconsistency), making calibration worse — the practical lesson being to calibrate with at least two points where possible.

7.5 The Pyrolysis Advisor

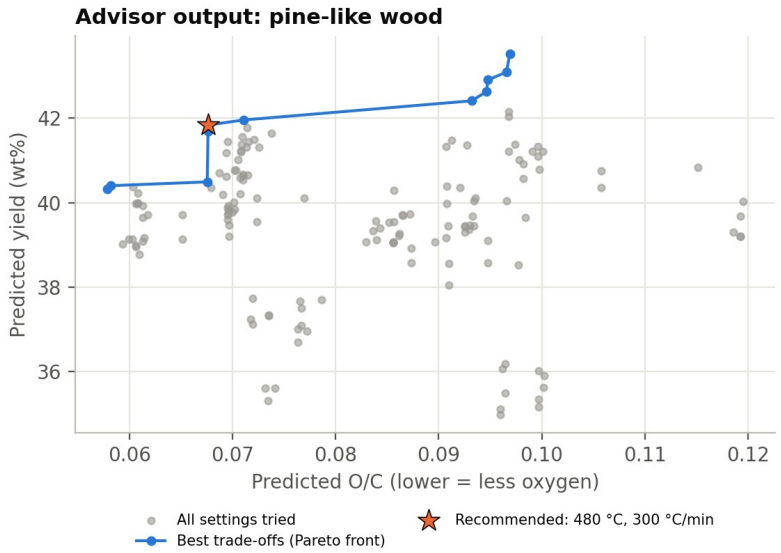


Figure 7.4: Advisor output for a pine-like wood.

Table 7.4: Advisor recommendations for four example feedstocks

Feedstock	Known?	Recommended	Alternative (max yield)
Pine-like wood	Yes	480°C, 300°C/min → 41.8 wt %, O/C 0.068	Max yield: 570°C, 100°C/min → 43.5 wt%, O/C 0.097
Rice husk-like	Yes	510°C, 10°C/min → 41.6 wt %, O/C 0.062	Max yield: 570°C, 100°C/min → 44.7 wt%, O/C 0.113
Microalgae-like (N 8.5%)	No: flagged	Shown with a warning	Needs 1–2 experiments first
Coal-like (C 80%)	No: flagged	Shown with a warning	Outside biomass composition range

The Pareto front makes the trade-off explicit: for pine, the last 1.7 wt% of yield costs ~40% more oxygen. The applicability check correctly withholds confident recommendations for the nitrogen-rich microalgae-like and carbon-rich coal-like feedstocks, both well outside either training dataset's composition range.

CHAPTER 8: SUPPORTING ENGINEERING — A DEPLOYABLE PREDICTION TOOL

Alongside the honest-benchmark study (Chapters 6–7), a separate full-stack application was built to make an earlier iteration of the O/C model usable outside a notebook: a FastAPI backend serving trained models via REST, and a React/TypeScript frontend for real-time prediction, batch scoring, model comparison and SHAP-based explanation.

8.1 Independent Physics Layer

This earlier iteration developed its own physics-informed features — distinct in formulation from Chapter 4 but built on the same principle: Arrhenius kinetics (independent parallel reactions) per pseudo-component; a competitive Broido-Shafizadeh cellulose pathway; a secondary tar-cracking severity index; an ash-catalysed devolatilisation correction; and the Modified Dulong HHV formula.

8.2 REST API

Key endpoints: POST /predict (O/C, confidence interval, secondary properties, physics-validation flags); POST /batch-predict (bulk CSV/Excel scoring); POST /explain (per-prediction SHAP attribution in plain chemistry language); POST /drift-check (out-of-range input flagging); POST /compare-models (all seven trained model types side by side).

8.3 Reported Accuracy and Reconciliation with Part A

Under an 80/20 random split, the best model (XGBoost) achieved O/C test R^2 = 0.900, versus 0.886 without physics features, and close to Zhang et al.'s own 0.895. Read against Chapter 6, this is a random-split figure of exactly the kind shown there to be optimistic: D1b's 122 biomasses mean an 80/20 row split very likely places the same biomass in both partitions. The grouped-by-biomass protocol of Chapter 5 is the more defensible standard this model should ultimately be re-validated against (listed as future work, Chapter 9); this application is presented as a deployment-engineering demonstration, not a second competing accuracy claim.

CHAPTER 9: CONCLUSION AND FUTURE WORK

9.1 Conclusion

Published ML accuracy for bio-oil pyrolysis is substantially an artefact of random testing: honestly tested on new biomasses, yield prediction drops to guessing-level and O/C to $R^2 \approx 0.2–0.3$ (Chapter 6). Hidden near-duplicates in compiled literature data make this worse unless explicitly clustered (Section 3.2). Physics-informed features, physical constraints, and a level-plus-shape design produce models that are honestly validated, follow known physics (Section 7.2), report realistic uncertainty, and improve substantially with 1–2 calibration experiments (Section 7.3). The resulting Pyrolysis Advisor gives practical, trade-off-aware guidance for more bio-oil with less oxygen, directly supporting the separation strategies in Dr. Akhil Mohan's review (Section 1.1).

9.2 Relation to Dr. Mohan's Separation Review

- Lower-O/C bio-oil reduces the solvent, adsorbent and energy demand of the separation stage; the Advisor (Section 7.5) makes this upstream choice explicit.
- Section 8.4 of the review calls for interpretable, physics-informed ML; this project supplies it and quantifies why purely data-driven literature models often fail to transfer (Chapter 2).
- Figure 11 of the review proposes a data → clean → feature → train → evaluate → validate → recommend pipeline, implemented here end to end including the honest validation step Chapter 2 shows is often skipped.

9.3 Limitations

- Compiled literature data mixes labs and measurement conventions; residence time, condensation conditions and product water content are not consistently reported.
- O/C and yield come from two separately trained models (no dataset reports both); the Advisor combines them rather than one jointly trained model.

- Uncertainty for a genuinely new biomass remains wide; the models are most useful refined by 1–2 real experiments, not as a replacement for them.
- No separation-stage dataset exists in the surveyed literature, so that stage is discussed but not modelled.
- The supporting web app (Chapter 8) is validated only by random split and not yet re-evaluated grouped-by-biomass.

9.4 Future Work

- Recover a source-study identifier and test leave-one-study-out validation, isolating the inter-lab effect (Section 6.3).
- Model organic yield (bio-liquid minus water) on the 413 D3 rows reporting water content.
- Collect internally consistent single-reactor data recording residence time and condensation temperature, and retrain.
- Extend the Advisor to the separation stage once suitable data exists.
- Re-validate the supporting web app's model under the grouped-by-biomass protocol of Chapter 5.

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APPENDIX 1: PROJECT FILE STRUCTURE

Per the report-format guidelines, complete source code is not reproduced in the body of this report; the full project (code, cleaned data, trained models, results, figures) is provided separately on CD/DVD or the accompanying repository, structured as follows.

Folder / file	Contents
data_raw/, data_clean/	Original and cleaned datasets with assigned biomass (FeedID) groups
src/01_clean.py, features.py, common.py	Cleaning, physics-feature functions, shared model/split utilities
src/02_leakage_study.py, 04_figures_A.py	Part A: honest benchmark and its figures (Chapter 6)
src/05, 06	Model-selection studies for the final O/C and yield models (Chapter 7)
src/07_final_advisor.py, 08_figures_B.py	Part B: final models, calibration, case studies, Advisor, figures (Chapter 7)
results/, models/	All result tables, figures and serialized final models